



TEAM NAME : BIT LORDS

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PROBLEM STATEMENT

PREDICTION OF PARKINSON'S DISEASE

- Parkinson's disease is a progressive neurological disorder that primarily affects movement.
- It effects the brain cells and with the help of this Parkinson's data set we can detect symptoms of neurodegenerative diseases in early stage.

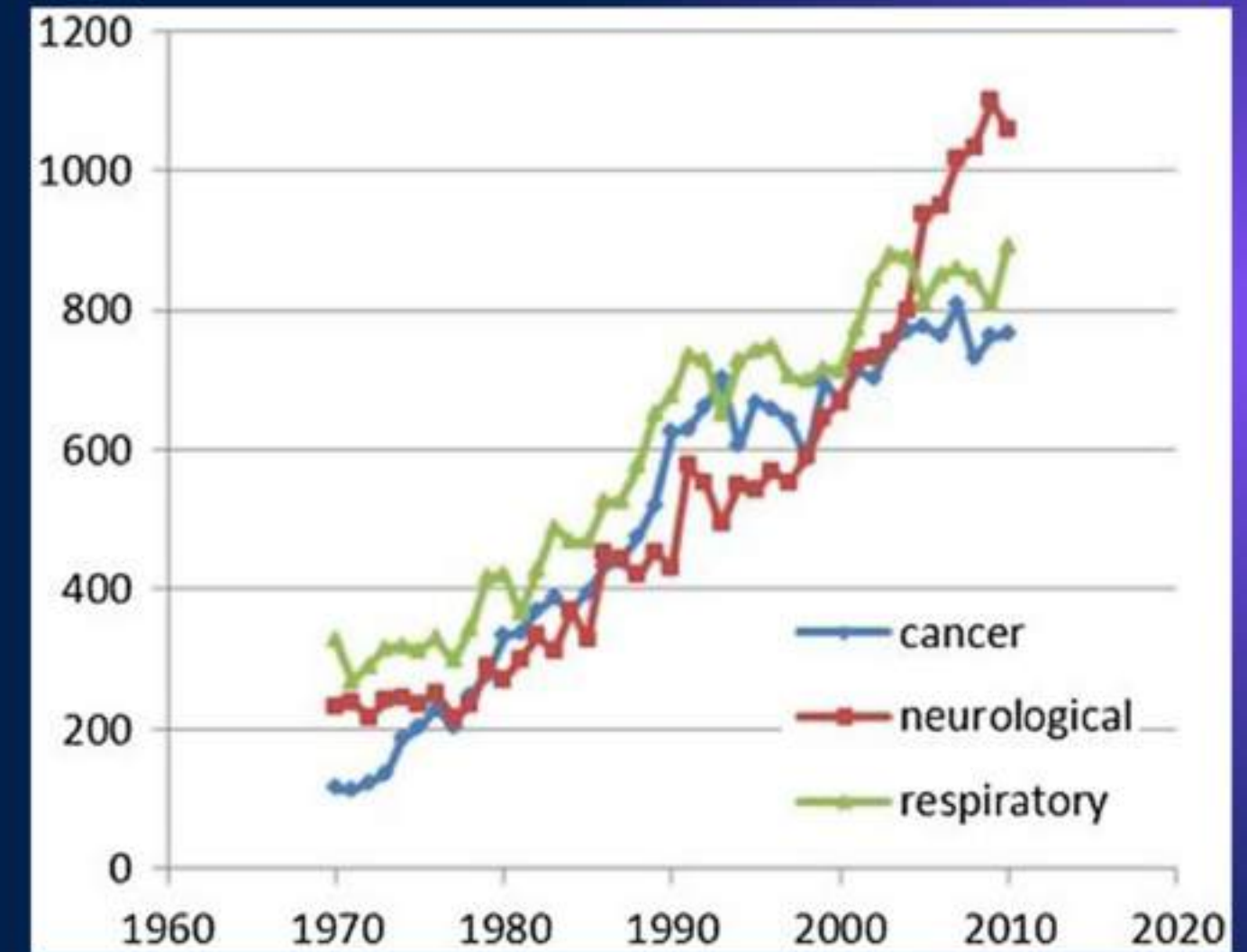


SOLUTIONS

- The solution provides the machine learning model for the Parkinson's disease .
 - A machine learning model that analyzes early signs of Parkinson's disease using key features like voice data, motor functions, and handwriting.
- It provides a cost-effective, non-invasive early detection method.
 - Preventing the neurological disorder and by early detection of the disease and taking further precautions
- Allowing the access for the poor ones who are unable to afford expensive test for such disorders
 - The model uses the voice pitch and frequency for the detection of the disorder

USP

- **Affordable & Accessible:** Uses simple, non-invasive data like voice recordings.
- **Early Detection:** Helps detect Parkinson's before symptoms become severe.
- **Scalability:** Can be implemented on a wide scale using mobile or wearable technology.
- **User-Friendly:** Designed for easy integration into healthcare systems.



TECH STACK

Front-end: Framework or tools used for the frontend

- **UI/UX** : For the designing the frontend view.
- **HTML** : Creating the Frontend of the model .
- **CSS** : Used for the creating the attractive application

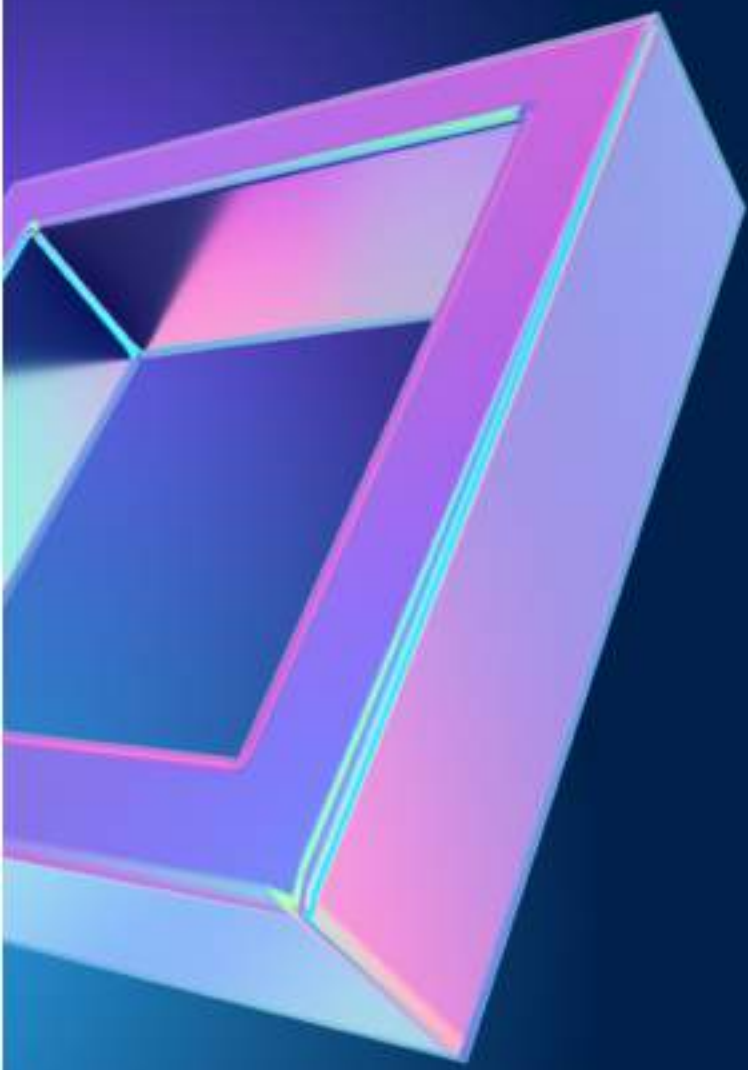
Machine Learning

- Random Forest:
Explain that it is a powerful classification algorithm used for Parkinson's detection because it handles noisy data well and prevents overfitting.

- Support Vector Machine (SVM): Used for separating data into classes, which makes it effective for detecting patterns in medical data like voice features and movement.
- K-Nearest Neighbors (KNN): Helps classify whether the patient has Parkinson's by comparing symptoms with similar cases.

FUTURE ASPECT :

- **Integration with Wearable Tech:** Use smartwatches to collect real-time motor data.
- **Data Expansion:** Incorporate MRI or brain imaging data for higher accuracy.
- **Global Scalability:** Deploying the model across different regions to assist in widespread screening.
- **Mobile App Development:** Creating a mobile platform for users to access the model directly.
- **Physical Dataset:** Collecting Physical DataSet such as Real time voice scan, handwriting scan, body gesture movement



THANK YOU